KINETIC THEORY OF GASES

FORMULA SHEET | JEE MAINS | Let's Play With Physics

1. IDEAL GAS EQUATION

$$PV = nRT = Nk_BT, \qquad P = \frac{\rho RT}{M}$$

$R = 8.314$ J mol^{-1}K^{-1}, $k_B = \dfrac{R}{N_A} = 1.38 \times 10^{-23}$ J/K

$N_A = 6.022 \times 10^{23}$ mol^{-1}; 1 mol at STP occupies 22.4 L

Gas laws: Boyle (PV = const at const T), Charles ($V \propto T$ at const P), Gay-Lussac ($P \propto T$ at const V), Avogadro (equal V, P, $T \Rightarrow$ equal N)

Dalton: $P = P_1 + P_2 + \ldots$ (non-reacting mixture)

⋆ Common Pitfall

- Always use absolute temperature (K)
- M in kg/mol when using SI ($M_{O_2} = 0.032$ kg/mol)

2. KINETIC THEORY: PRESSURE

Assumptions: point molecules, elastic collisions, no intermolecular force (except during collision), random motion, negligible collision time

$$P = \frac{1}{3}\frac{mN}{V}\overline{v^2} = \frac{1}{3}\rho v_{\text{rms}}^2$$

$$P = \frac{2}{3}E, \quad E = \text{translational KE per unit volume}$$

⋆ Common Pitfall

- Pressure depends on $\overline{v^2}$, not $\bar{v}^2$
- Collision with the wall reverses only the normal velocity component: $\Delta p = 2mv_x$

3. MOLECULAR SPEEDS

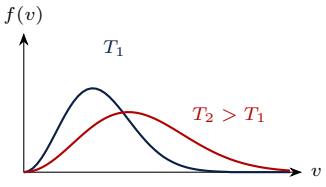
$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3k_BT}{m}} = \sqrt{\frac{3P}{\rho}}$$

$$v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}, \qquad v_{\text{mp}} = \sqrt{\frac{2RT}{M}}$$

$$v_{\text{rms}} : v_{\text{avg}} : v_{\text{mp}} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.73 : 1.60 : 1.41$$

Speed of sound: $v_s = \sqrt{\dfrac{\gamma RT}{M}}$, $\dfrac{v_{\text{rms}}}{v_s} = \sqrt{\dfrac{3}{\gamma}}$

Escape: gas escapes a planet when $v_{\text{rms}} \gtrsim v_{\text{esc}}$ (fraction escapes earlier)



Maxwell distribution: area under curve = N (constant); higher $T \Rightarrow$ peak shifts right and lowers

⋆ Common Pitfall

- $v_{\text{rms}} \propto \sqrt{T/M}$; independent of P at constant T
- Order: $v_{\text{rms}} > v_{\text{avg}} > v_{\text{mp}}$
- rms speed of a set of numbers $= \sqrt{\frac{\sum v_i^2}{N}}$, not the mean

4. KINETIC ENERGY & TEMPERATURE

Avg translational KE per molecule: $\frac{3}{2}k_BT$

Translational KE per mole: $\frac{3}{2}RT$

Total KE (internal energy) of n moles: $U = \frac{f}{2}nRT$

⋆ Common Pitfall

- Translational KE per molecule depends only on T, same for all gases
- Absolute zero: translational KE of ideal gas molecules would be zero

5. DEGREES OF FREEDOM & EQUIPARTITION

Law of equipartition: each quadratic degree of freedom gets $\frac{1}{2}k_BT$ per molecule

Gas	f	C_V	C_P	γ
Monatomic (He, Ar)	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} = 1.67$
Diatomic rigid (O_2, N_2)	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$
Diatomic + vibration	7	$\frac{7}{2}R$	$\frac{9}{2}R$	$\frac{9}{7} = 1.29$
Polyatomic non-linear	6	$3R$	$4R$	$\frac{4}{3} = 1.33$

$$C_V = \frac{f}{2}R, \quad C_P = C_V + R, \quad \gamma = \frac{C_P}{C_V} = 1 + \frac{2}{f}$$

$$f = 3N - k \quad (N = \text{atoms}, \ k = \text{constraints})$$

Vibrational mode contributes $2 \times \frac{1}{2}k_BT = k_BT$ (KE + PE)

Solids (Dulong–Petit): $C = 3R$

⋆ Common Pitfall

- Diatomic: $f = 5$ at room temperature (vibration frozen); $f = 7$ only at high T
- Linear polyatomic (CO_2, rigid) has $f = 5$, non-linear has $f = 6$
- $C_P - C_V = R$ (Mayer's relation) for ideal gases only

6. GAS MIXTURES

$$C_{V,\text{mix}} = \frac{n_1C_{V1} + n_2C_{V2}}{n_1 + n_2}$$

$$\frac{n_1 + n_2}{\gamma_{\text{mix}} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

$$M_{\text{mix}} = \frac{n_1M_1 + n_2M_2}{n_1 + n_2}$$

Mixing at different temperatures (insulated): $T_{\text{mix}} = \dfrac{n_1f_1T_1 + n_2f_2T_2}{n_1f_1 + n_2f_2}$

⋆ Common Pitfall

- Never average γ directly — average C_V, then $\gamma = 1 + R/C_V$

7. MEAN FREE PATH

$$\lambda = \frac{1}{\sqrt{2}\,\pi d^2\, n} = \frac{k_B T}{\sqrt{2}\,\pi d^2 P}$$

$n = N/V$ (number density), d = molecular diameter

Collision frequency $= \dfrac{v_{\text{avg}}}{\lambda}$

⋆ Common Pitfall

- $\lambda \propto T/P$; at constant density, λ is independent of T
- $\sqrt{2}$ comes from relative motion of molecules

8. REAL GASES

van der Waals:

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

a: intermolecular attraction; b: finite molecular volume

Real gases behave ideally at **low pressure** and **high temperature**

9. JEE SHORTCUTS & MASTER TIPS

1. $v_{\text{rms}} = \sqrt{3RT/M}$; same T: $v \propto 1/\sqrt{M}$
2. $P = \frac{1}{3}\rho v_{\text{rms}}^2$ and $P = \frac{2}{3}E$
3. $\gamma = 1 + 2/f$; $C_V = fR/2$
4. KE per molecule $= \frac{3}{2}k_B T$; $U = \frac{f}{2}nRT$
5. Mixture: $\dfrac{\sum n}{\gamma_{\text{mix}} - 1} = \sum \dfrac{n_i}{\gamma_i - 1}$
6. $\lambda \propto \dfrac{T}{Pd^2}$

⋆ Master Confusion List

1. **Translational vs total KE:** $\frac{3}{2}k_B T$ vs $\frac{f}{2}k_B T$ per molecule
2. k_B **vs** R**:** per molecule vs per mole
3. f **for diatomic:** 5 normally, 7 only with vibration
4. **Speeds:** rms > avg > mp
5. **Mixture** γ**:** don't average γ values
6. **Mean free path** depends on n, not directly on T

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